Maths

Probability

1. Key concepts in probability theory:
    1. Let's say we have two RV X and Y:
        1. Joint distribution: $p(X, Y)$ (reads "X and Y")
        2. Marginal distribution: p(X) and p(Y)
        3. Conditional distribution: p(X|Y) and p(Y|X) (reads "X given Y" and "Y given X")
        4. Bayes Theorem: $p(X|Y) = \frac{p(X,Y)}{p(Y)} = \frac{p(Y|X)p(X)}{p(Y)}$
            1. Prior: p(X) (what I originally believe about X)
            2. Posterior: p(X|Y) (what I believe about X now that I've seen/observed Y).
        5. Independence: Knowing X tells you nothing about Y
            1. $p(X|Y) = p(X)$ and $p(X, Y) = p(X)p(Y)$.
    2. The goal of probabilistic modeling is to learn the probability distributions
    3. We can usually describe the probability distributions through some **parameters** (θ)
        1. Gaussian: mean(μ) and variance(σ^2)
        2. Poisson: average rate (λ)
        3. Some complicated distribution: can be parametrized by neural networks (θ)
    4. The goal now is to learn those parameters given data
    5. We call the probability of data given model parameters as **likelihood** $p(x|\theta)$. i.e. "if my parameters are accurate, how likely is the data that I observe".
        1. θ is model parameter.
        2. For example RV x is binomial, x is the number of times a coin landed heads in N trials.
            1. $$P(x = X|\theta = p) = \binom{N}{x} p^x (1-p)^{N-x} \quad \text{(likelihood)}$$
        3. if RV x is single variable Gaussian, for modeling a real value, e.g. temperature sensor reading.
            1. $$P(x = X|\theta = (\mu, \sigma^2)) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad \text{(likelihood)}$$
        4. Note: Notice that in both cases the RV x ended up in the exponent.
        5. Bayes rule:

$$p(\theta|x = D) = \frac{p(x|\theta)p(\theta)}{p(x)}$$

$p(x|\theta)$ is the likelihood, $p(\theta)$ is the prior distribution, $p(\theta|x = D)$ is the posterior, and $p(x)$ is the normalization term.

2. Probabilistic graphical models can only do 2 things [4]:
    1. Inference: calculating a probability of some data modelled as RV using the model $\mathcal{M}$:
        1. evaluation $P(obs|state)$,
        2. decoding $P(state|obs)$
    2. Learning model parameters: refers to finding the graphical model $\mathcal{M}$, given some data $\mathcal{D}$.
        1. MAP (maximum a posteriori)
        2. Maximum likelihood
        3. expectation maximization
        4. forward-backward algorithm
3. Conditional probability can be computed using:
    1. Bayes rule
    2. Enumeration

Probabilistic Inference Methods [5]

1. Maximum Likelihood Estimation (MLE)
    1. $x^* = \arg\max p(d|x)$.

    2. Likelihood: $p(d|x)$
    3. $x*$ is the most probable value of x.
    4. $\{x\}$ is a set of possible values of x (the candidates of x).
    5. $\arg\max$ to return the value of x that maximizes $p(d|x)$.
    6. The weakness of MLE is that it allows impossible candidates, as it treats candidates uniformly.
    7. The intuitive meaning of MLE:
        1. Take each possible illness e.g. x=flu.
        2. Verify if it fits to the symptoms $\rightarrow$ probability of x.
        3. Decide the illness based on the hights probability.
2. Maximum A Posteriori (MAP)
    1. $$\begin{aligned} x^* &= \arg\max p(x|d) \\ &= \arg\max p(d|x)p(x) \end{aligned}$$
       . Likelihood:
    2. MAP treats the candidates differently: $p(x = \text{flu}) > p(x = \text{cancer})$
    3. Main criticism of MAP is how to determine $p(x)$.
3. Full Bayesian Inference
    1. $p(x|d) = \frac{p(d|x)p(x)}{p(d)}$. $P(d)$ is intractable to solve.
    2. This is the ideal and most robust inference that besides providing a prediction on the value of x, it also provides the uncertainty, or how high/low the probability of the prediction.
    3. It's possible because of $p(d)$, the evidence. This will normalize the product of $p(d|x)p(x)$.
    4. However to obtain $p(d)$ is analytically intractable, and in discrete, it can be prohibitively expensive.
4. Example: According to our observation, a patient has a set of symptoms: d : {stiff neck, coughing, headache, fever}. We also have a set of possible illness: flu, meningitis, infection and cancer.

Lagrange Multiplier [3]

Hungarian Matching [2]

AM, GM [8]

1. **The geometric mean of two numbers is always less than their arithmetic mean**. The two are equal only if x and y are equal.
    1. Proof: Let x and y be the dimensions of rectangle in inches.
    2. It can be proven that **the square is larger than all other rectangles with the same perimeter**
    3. The are of the rectangle may then be given in square inches and its perimeter is $x + y + x + y = 2(x + y)$ inches. Therefore the square has sides $\frac{1}{2}(x + y)$ inches in length.
    4. Thus if x and y are any two positive numbers, the results becomes: $xy \leq \left(\frac{x+y}{2}\right)^2$.
    5.
2. The arithmetic mean is not larger than the root mean square.
3. The geometric mean of n positive numbers is never larger than the arithmetic mean, $G \leq A$.

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